SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4 **Total Marks:40**

Annexure A

CASE STUDY

Code of Conduct:

I ***Devadharshini E(192524483)*** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Devadharshini E*

Annexure A

CASE STUDY

Case Study 1 — Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

I. If a patient has Fever AND Cough → Possible Flu

Ii. If a patient has Fever AND Rash → Possible Measles

Iii. If a patient has Cough AND Breathlessness → Possible Pneumonia

Iv. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

(a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.

(b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.

(c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Case Study 2 — Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

I. ∀x: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x) [Rule 1]

II. ∀x: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x) [Rule 2]

III. ∀x ∀y: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x) [Rule 3]

IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

(a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.

(b) Using the Resolution method, prove that 'Proceed (CarA)' is true. Convert relevant clauses to CNF and show each resolution step.

(c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Case Study 3 — Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

1) P1: SoilDry → IrrigationNeeded
2) P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod
3) P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk
4) Facts: SoilDry = True, CropWheat = True

(a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication → move ¬ inward → distribute).

(b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause □. Draw the resolution proof tree.

(c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Case Study 4 — Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

1) Stench [1,2] → Wumpus is adjacent to cell [1,2]
2) Breeze [1,1] → Pit is adjacent to cell [1,1]

3) Glitter [x,y] → Gold is at [x,y]

4) ¬Wumpus [x,y] ∧ ¬Pit[x,y] → Safe[x,y]

(a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.

(b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.

(c) Evaluate whether the path [1,1] → [1,2] → [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Question 1:-

SMART MEDICAL DIAGNOSIS SYSTEM

(a) Propositional Logic Representation

Let:

- F = Patient A has Fever
- C = Patient A has Cough
- R = Patient A has Rash
- B = Patient A has Breathlessness
- Flu = Patient A has Possible Flu
- M = Patient A has Possible Measles
- P = Patient A has Possible Pneumonia

Rules

1. Fever AND Cough → Possible Flu

$$F \land C \rightarrow FL$$

2. Fever AND Rash → Possible Measles

$$F \land R \rightarrow M$$

3. Cough AND Breathlessness → Possible Pneumonia

$$C \land B \rightarrow P$$

Facts

F = True

C = True

B = True

There is no fact stating that Rash is true, so is unknown.

(b) Forward Chaining starts with the known facts and applies the rules whose conditions are satisfied.

Initial Facts

F = True

C = True

B = True

Rule I : $F \land C \rightarrow FL$

Given: F = True, C = True

Therefore: $F \land C \rightarrow FL$, so FL = True

Derived Fact: Patient A has Possible Flu.

Rule II: $F \land R \rightarrow M$

Given: F = True

However, R = Rash is not given as True.

Therefore, Rule II cannot be fired.

Derived Fact: Possible Measles cannot be concluded.

Rule III: $C \land B \rightarrow P$

Given: C = True, B = True

Therefore: $C \land B \rightarrow P$, so P = True

Derived Fact: Patient A has Possible Pneumonia.

(c) Backward Chaining for Pneumonia

Goal: Patient A has Possible Pneumonia.

Backward Chaining starts from the goal and searches for a rule that can derive it.

Goal Reduction Tree

```
                Goal: Pneumonia (P)
                        |
               Rule III: C ∧ B → P
                     /      \
                    /        \
          Cough (C)      Breathlessness (B)
                  |              |
               TRUE           TRUE
```

Both required conditions are already available as facts:

C = True

B = True

Therefore: $C \land B \rightarrow P$

P = True

Therefore, the goal “Patient A has Pneumonia” is successfully proved using Backward Chaining.

Question 2:-

AUTONOMOUS TRAFFIC MANAGEMENT AGENT

(a) Unification of Rule 1 with the Given Facts

Rule 1

$\forall x$: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)

For unification, the rule is applied to the given facts:

Vehicle(Ambulance)

EmergencyType(Ambulance)

Step 1 – Match Vehicle(x)

Rule predicate: Vehicle(x)

Fact: Vehicle(Ambulance)

Therefore: x = Ambulance

Substitution: $\theta_1 = \{x/Ambulance\}$

Step 2 – Match EmergencyType(x)

After applying θ_1: EmergencyType(Ambulance)

Given fact: EmergencyType(Ambulance)

Therefore, the second condition also matches.

Substitution remains: $\theta_2 = \{x/Ambulance\}$

Step 3 – Derive the Conclusion

Applying $\theta = \{x/Ambulance\}$ to Rule 1:

Vehicle(Ambulance) ∧ EmergencyType(Ambulance) → ClearPath(Ambulance)

Since both facts are true: ClearPath(Ambulance)

Final MGU

$\theta = \{x/Ambulance\}$

Thus, Rule 1 successfully matches the emergency vehicle Ambulance and derives:

ClearPath(Ambulance)

(b) Resolution Proof of Proceed(CarA)

Step 1 – Convert the Relevant Rules into CNF

Rule 1: Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)

CNF: ¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)

Rule 2: ClearPath(x) → GreenSignal(x) ∧ ¬RedSignal(x)

This gives two clauses:

¬ClearPath(x) ∨ GreenSignal(x)

¬ClearPath(x) ∨ ¬RedSignal(x)

Rule 3: Vehicle(x) ∧ Behind(x,y) ∧ GreenSignal(y) → Proceed(x)

CNF: ¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)

Given Facts

Vehicle(Ambulance)

EmergencyType(Ambulance)

Vehicle(CarA)

Behind(CarA, Ambulance)

Step 2 – Derive ClearPath(Ambulance)

From Rule 1: ¬Vehicle(x) ∨ ¬EmergencyType(x) ∨ ClearPath(x)

Using: Vehicle(Ambulance)

and: EmergencyType(Ambulance)

Resolution gives: ClearPath(Ambulance)

Step 3 – Derive GreenSignal(Ambulance)

From Rule 2: ¬ClearPath(x) ∨ GreenSignal(x)

Using: ClearPath(Ambulance)

Resolution gives: GreenSignal(Ambulance)

Step 4 – Derive Proceed(CarA)

Rule 3: ¬Vehicle(x) ∨ ¬Behind(x,y) ∨ ¬GreenSignal(y) ∨ Proceed(x)

Substitute:

x = CarA

y = Ambulance

We get: ¬Vehicle(CarA) ∨ ¬Behind(CarA,Ambulance) ∨ ¬GreenSignal(Ambulance) ∨ Proceed(CarA)

Using the facts:

Vehicle(CarA)

Behind(CarA,Ambulance)

GreenSignal(Ambulance)

Resolution eliminates the first three negative literals and gives:

Proceed(CarA)

Resolution Chain

Vehicle(Ambulance)

+

EmergencyType(Ambulance)

↓

ClearPath(Ambulance)

↓

GreenSignal(Ambulance)

↓

Vehicle(CarA) + Behind(CarA,Ambulance)

↓

Proceed(CarA)

Therefore:

Proceed(CarA) is proved to be true.

(c) Handling Two Simultaneous Emergency Vehicles

The current knowledge base is not completely sufficient to handle two simultaneous emergency vehicles.

The existing rules use a general variable x:

Vehicle(x) ∧ EmergencyType(x) → ClearPath(x)

Therefore, the rule can identify more than one emergency vehicle if the corresponding facts are provided. For example:

Vehicle(Ambulance1)

EmergencyType(Ambulance1)

Vehicle(Ambulance2)

EmergencyType(Ambulance2)

The rule can derive:

ClearPath(Ambulance1)

ClearPath(Ambulance2)

However, the knowledge base does not contain rules for priority, conflict resolution, or simultaneous emergency routing.

Additional Facts Needed

For example:

Vehicle(Ambulance1)

EmergencyType(Ambulance1)

Vehicle(Ambulance2)

EmergencyType(Ambulance2)

Their locations or relationships should also be represented, such as:

$$At(Ambulance1,Road1)$$

$$At(Ambulance2,Road2)$$

Additional Rules Needed

A priority rule can be added:

$$EmergencyType(x) \land HigherPriority(x,y) \rightarrow GivePriority(x)$$

A conflict-handling rule can also be added:

$$EmergencyVehicle(x) \land EmergencyVehicle(y) \land SameRoute(x,y) \rightarrow ManageConflict(x,y)$$

These rules allow the system to determine which emergency vehicle should receive priority when both require the same route or signal.

Question 3:-

<u>AGRICULTURAL EXPERT REASONING SYSTEM</u>

(a) Conversion into Conjunctive Normal Form (CNF)

Given:

$$P1: SoilDry \rightarrow IrrigationNeeded$$

$$P2: IrrigationNeeded \land CropWheat \rightarrow ApplyDripMethod$$

$$P3: \neg ApplyDripMethod \land CropWheat \rightarrow CropAtRisk$$

Facts:

$$SoilDry = True$$

$$CropWheat = True$$

P1: SoilDry → IrrigationNeeded

Eliminate implication: ¬SoilDry ∨ IrrigationNeeded

This is already in CNF.

Therefore: P1 = ¬SoilDry ∨ IrrigationNeeded

P2: IrrigationNeeded ∧ CropWheat → ApplyDripMethod

Eliminate implication:

$$\neg(IrrigationNeeded \land CropWheat) \lor ApplyDripMethod$$

Move ¬ inward using De Morgan's law:

$$\neg IrrigationNeeded \lor \neg CropWheat \lor ApplyDripMethod$$

This is already in CNF.

Therefore:

$$P2 = \neg IrrigationNeeded \lor \neg CropWheat \lor ApplyDripMethod$$

P3: ¬ApplyDripMethod ∧ CropWheat → CropAtRisk

Eliminate implication:

$$\neg(\neg \text{ApplyDripMethod} \land \text{CropWheat}) \lor \text{CropAtRisk}$$

Move ¬ inward:

$$\text{ApplyDripMethod} \lor \neg \text{CropWheat} \lor \text{CropAtRisk}$$

This is already in CNF.

Therefore:

$$P3 = \text{ApplyDripMethod} \lor \neg \text{CropWheat} \lor \text{CropAtRisk}$$

Facts in CNF

$$\text{SoilDry} = \text{True}$$

$$\text{SoilDry}$$

$$\text{CropWheat} = \text{True}$$

$$\text{CropWheat}$$

Final CNF Knowledge Base

1. ¬SoilDry ∨ IrrigationNeeded
2. ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod
3. ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk
4. SoilDry
5. CropWheat

(b) Resolution Refutation to Prove ApplyDripMethod

Goal: ApplyDripMethod

First, negate the goal: ¬ApplyDripMethod

Add this negated goal to the CNF knowledge base.

Step 1

From: ¬SoilDry ∨ IrrigationNeeded

and: SoilDry

Resolve SoilDry: IrrigationNeeded

Step 2

From: ¬IrrigationNeeded ∨ ¬CropWheat ∨ ApplyDripMethod

and: IrrigationNeeded

Resolve IrrigationNeeded: ¬CropWheat ∨ ApplyDripMethod

Step 3

From: ¬CropWheat ∨ ApplyDripMethod

and: CropWheat

Resolve CropWheat: ApplyDripMethod

Step 4

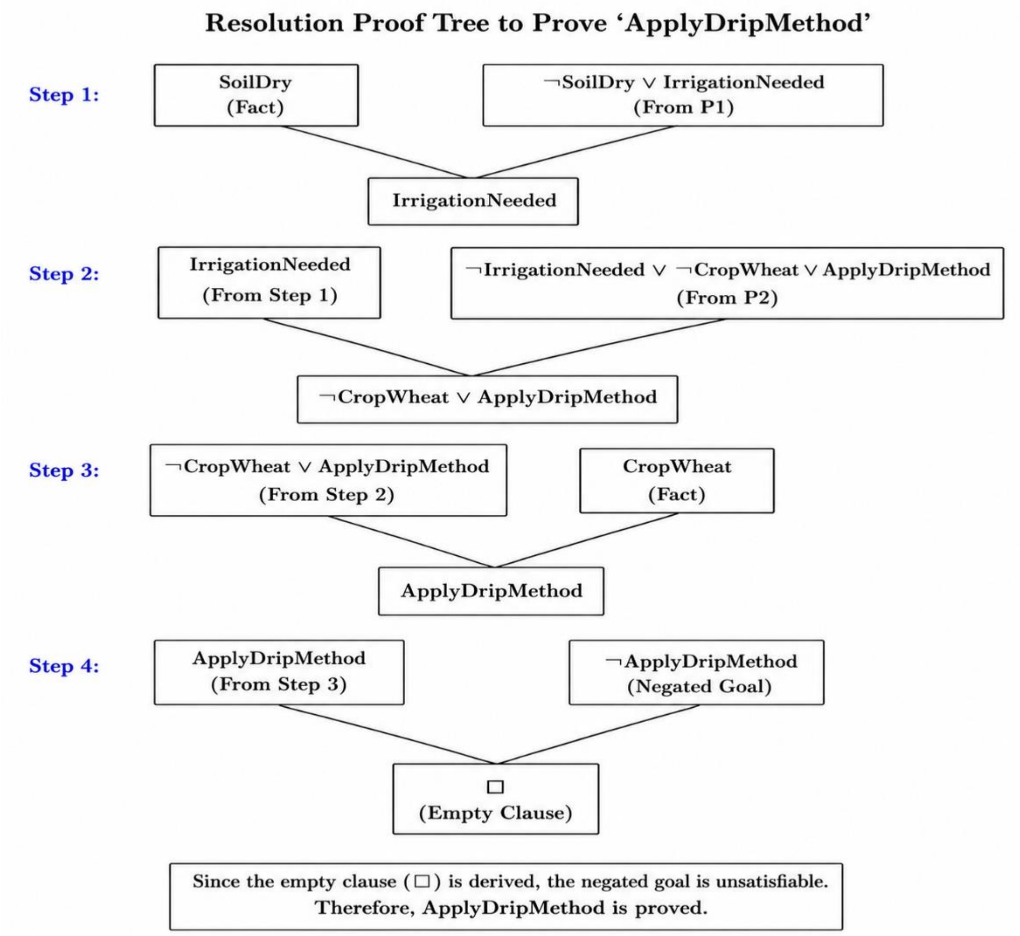
We now have: ApplyDripMethod

and the negated goal: ¬ApplyDripMethod

Resolving them gives: □

Therefore, the empty clause □ is obtained.

Hence, the theorem: ApplyDripMethod is proved.



(c) Removing SoilDry from the Knowledge Base

Suppose the fact: SoilDry is removed.

The remaining facts are: CropWheat = True

The rule: ¬SoilDry ∨ IrrigationNeeded

still exists, but there is no fact proving SoilDry.

Therefore, we cannot derive: IrrigationNeeded

Since IrrigationNeeded cannot be established, P2 cannot be used to derive: ApplyDripMethod

So the available information becomes: CropWheat

But there is also no fact proving: ¬ApplyDripMethod

Therefore, P3:

ApplyDripMethod ∨ ¬CropWheat ∨ CropAtRisk

cannot be used to conclude CropAtRisk because neither ¬ApplyDripMethod nor ApplyDripMethod is established.

Question 4:-

WUMPUS WORLD KNOWLEDGE AGENT

(a) Constructing the Belief State

The agent receives the following percepts:

- Stench at [1,2]
- Breeze at [1,1]
- Glitter at [2,2]

Using the given rules, the agent updates its belief state.

Step 1 – Stench at [1,2]

Rule 1:

Stench[1,2] → Wumpus is adjacent to [1,2]

Since Stench[1,2] is true:

Wumpus is adjacent to [1,2]

The possible adjacent cells are:

[1,1], [1,3], [2,2]

Therefore: Wumpus ∈ {[1,1], [1,3], [2,2]}

Step 2 – Breeze at [1,1]

Rule 2:

Breeze[1,1] → Pit is adjacent to [1,1]

Since Breeze[1,1] is true:

Pit is adjacent to [1,1]

The possible adjacent cells are:

[1,2], [2,1]

Therefore: Pit ∈ {[1,2], [2,1]}

Step 3 – Glitter at [2,2]

Rule 3:

Glitter[x,y] → Gold[x,y]

Since Glitter[2,2] is true:

Gold[2,2]

Therefore, gold is located at [2,2].

Step 4 – Safe Cells

Rule 4:

¬Wumpus[x,y] ∧ ¬Pit[x,y] → Safe[x,y]

The current percepts do not provide enough information to prove that any particular cell has neither a Wumpus nor a pit. Therefore, Safe[x,y] cannot be conclusively derived for the cells involved.

Derived Conclusions

1. Wumpus is adjacent to [1,2].
2. Wumpus may be at [1,1], [1,3], or [2,2].
3. Pit is adjacent to [1,1].
4. Pit may be at [1,2] or [2,1].
5. Gold is at [2,2].
6. No additional Safe cell can be conclusively derived from the given facts.

(b) Forward Chaining to Identify Possible Wumpus and Pit Locations

Forward Chaining starts with the known percepts and applies the rules whose conditions are satisfied.

Step 1

Known fact: Stench[1,2]

Apply Rule 1: Stench[1,2] → Wumpus is adjacent to [1,2]

Therefore: Wumpus ∈ {[1,1], [1,3], [2,2]}

Step 2

Known fact: Breeze[1,1]

Apply Rule 2: Breeze[1,1] → Pit is adjacent to [1,1]

Therefore: Pit ∈ {[1,2], [2,1]}

Step 3

Known fact: Glitter[2,2]

Apply Rule 3: Glitter[2,2] → Gold[2,2]

Therefore: Gold[2,2]

Final Belief

Wumpus possible locations: [1,1], [1,3], [2,2]

Pit possible locations: [1,2], [2,1]

Gold location:[2,2]

The available information does not allow the agent to determine the exact location of the Wumpus or Pit.

(c) Safety of the Path [1,1] → [1,2] → [2,2]

The proposed path is:

[1,1] → [1,2] → [2,2]

Cell [1,1]

The agent initially has a Breeze at [1,1]. This means a pit is present in one of its adjacent cells.

The possible pit locations are:

[1,2] or [2,1]

Therefore, [1,1] itself is not identified as containing a pit or Wumpus, but moving from it to [1,2] is risky.

Cell [1,2]

There is a Stench at [1,2], which means a Wumpus is adjacent to this cell.

Also, [1,2] is one of the possible pit locations because of the Breeze at [1,1].

Therefore, [1,2] cannot be confirmed as safe.

Cell [2,2]

Glitter[2,2] means:

Gold[2,2]

So the gold is definitely at [2,2].

However, [2,2] is also one of the possible locations of the Wumpus based on the Stench at [1,2].

Therefore, this cell cannot currently be confirmed as safe.

Final Conclusion

The path:

[1,1] → [1,2] → [2,2]

cannot be considered completely safe based on the available knowledge.